COMSATS University Islamabad
Department of Computer Science

Course Code: CSC262

Course Title: Artificial Intelligence Credit Hours: 3(2,1)

Lecture Hours/Week: 2

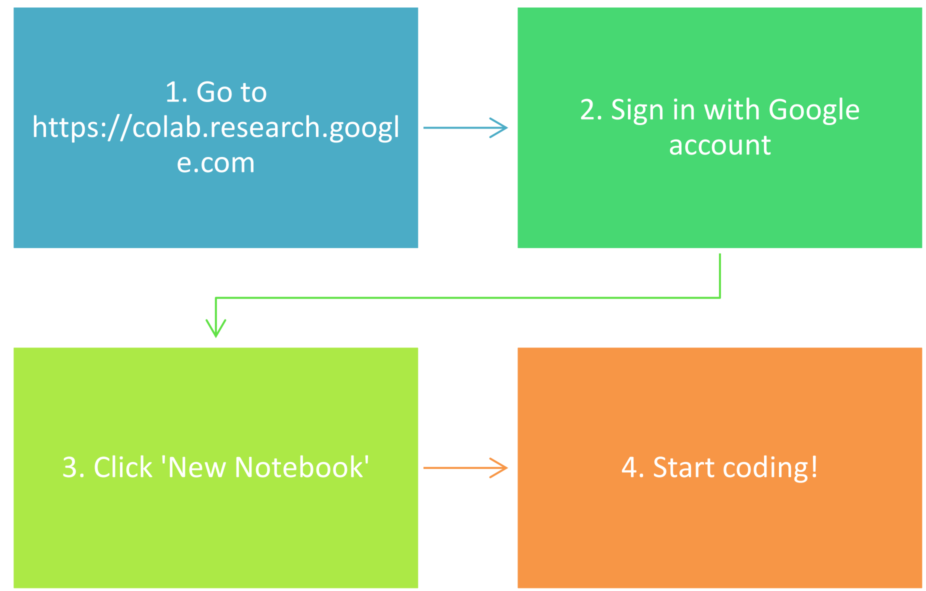
Lab Hours/Week: 3

Pre-Requisites: None

Why Use Google Colab?

Free access to computing resources	Easy sharing and collaboration	Ideal for Machine Learning & Data Science
Create/Upload/Share notebooks	Import/Save notebooks from/to Google Drive	Import/Publish notebooks from GitHub
Import external datasets e.g. from Kaggle	Integrate PyTorch, TensorFlow, Keras, OpenCV	Free Cloud service with free GPU

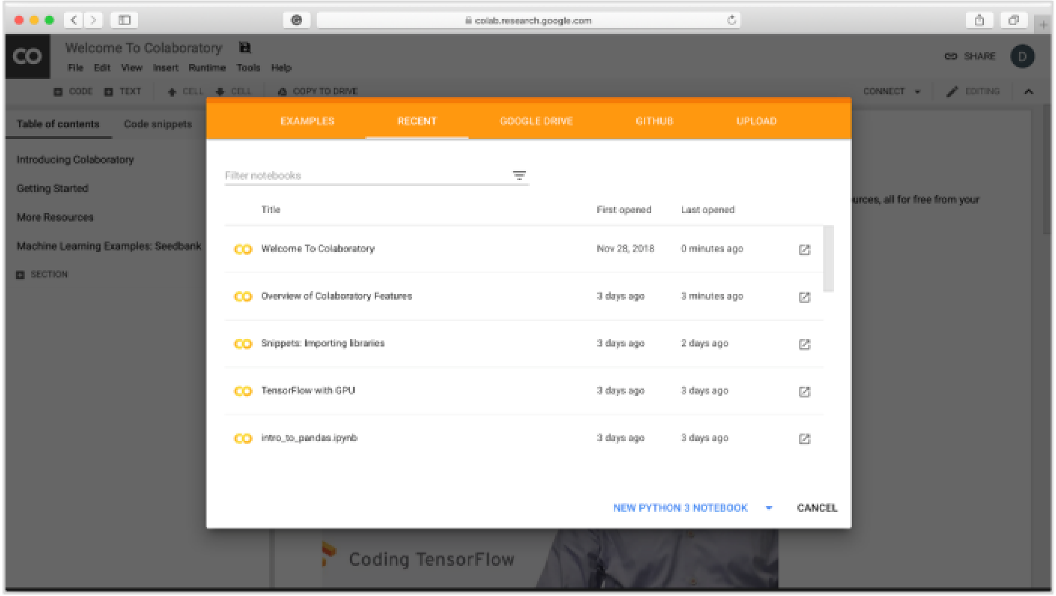
Getting Started



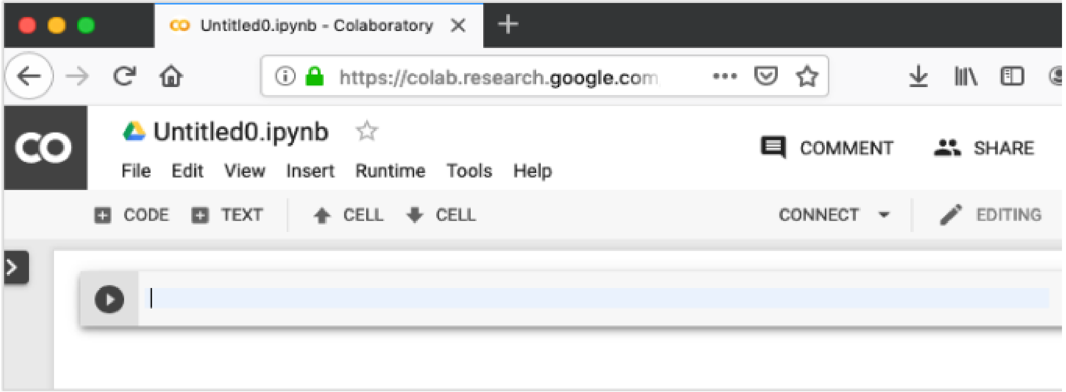
Colab Interface Overview

- Code cells: Write Python code
- Text cells: Add explanations (Markdown)
- Run button: Execute code
- Runtime menu: Manage resources

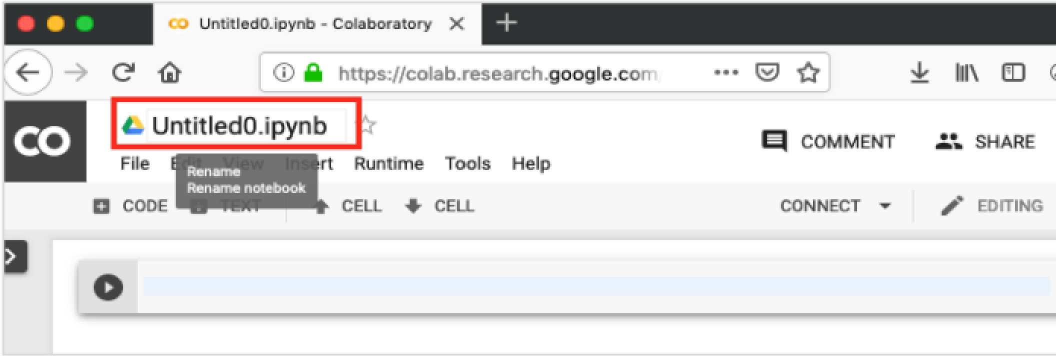
Your browser would display the following screen (assuming that you are logged into your Google Drive):



Click on the NEW PYTHON 3 NOTEBOOK link at the bottom of the screen. A new notebook would open up as shown in the screen below.

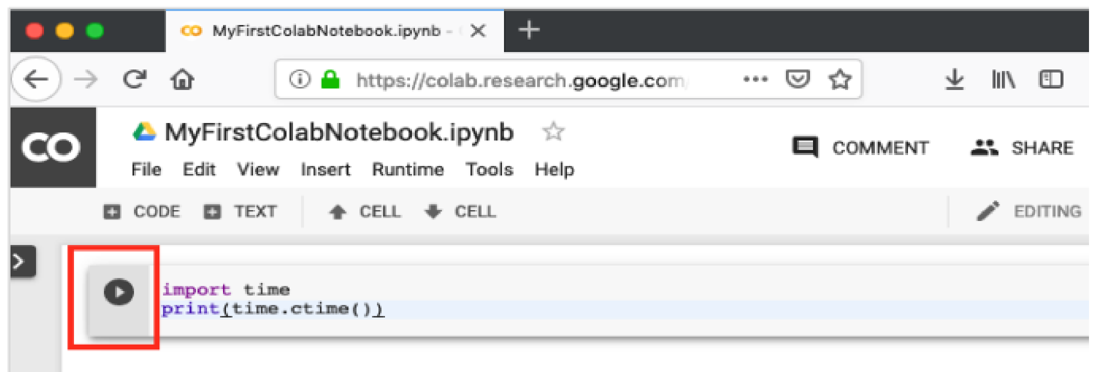


By default, the notebook uses the naming convention UntitledXX.ipynb. To rename the notebook, click on this name and type in the desired name in the edit box as shown here:

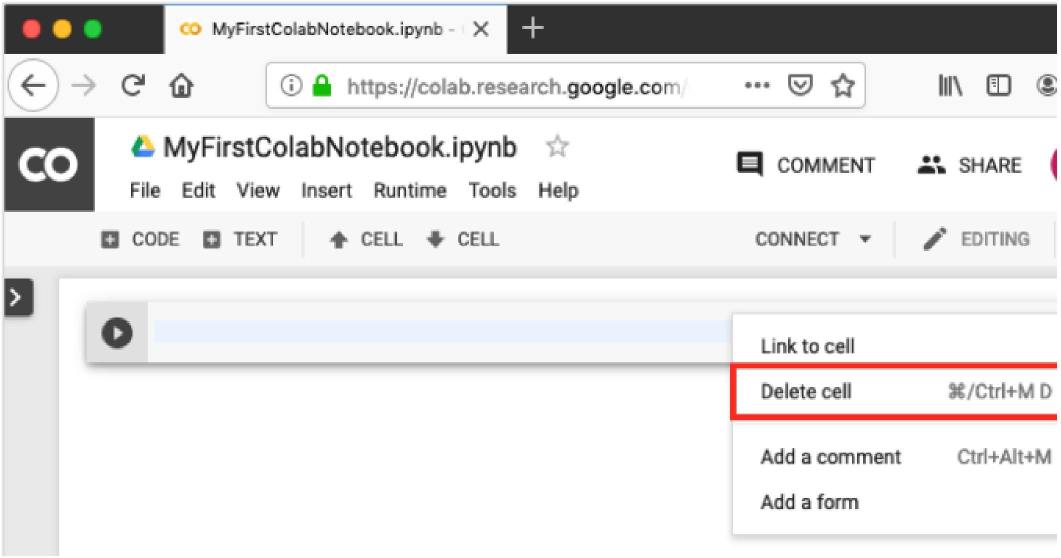


Entering Code

- You will now enter a trivial Python code in the code window and execute it. Enter the following two Python statements in the code window:
- import time
- print(time.ctime())



Deleting Cell



Google Colab – Documenting Your Code

- A text cell containing few mathematical equations typically used in ML is shown in the screenshot below:

Constraints are

- $3x_1 + 6x_2 + x_3 =< 28$
- $7x_1 + 3x_2 + 2x_3 =< 37$
- $4x_1 + 5x_2 + 2x_3 =< 19$
- $x_1, x_2, x_3 >= 0$

The trial vector is calculated as follows:

- $u_i(t) = x_i(t) + \beta(\hat{x}(t) - x_i(t)) + \beta \sum_{k=1}^{n_v}(x_{i1,k}(t) - x_{i2,k}(t))$

$$f(x_1, x_2) = 20 + e - 20exp(-0.2\sqrt{\frac{1}{n}(x_1^2 + x_2^2)}) - exp(\frac{1}{n}(cos(2\pi x_1) + cos(2\pi x_2)) \; x \in [-5, 5]$$

$$A_{m,n} = \begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{pmatrix}$$

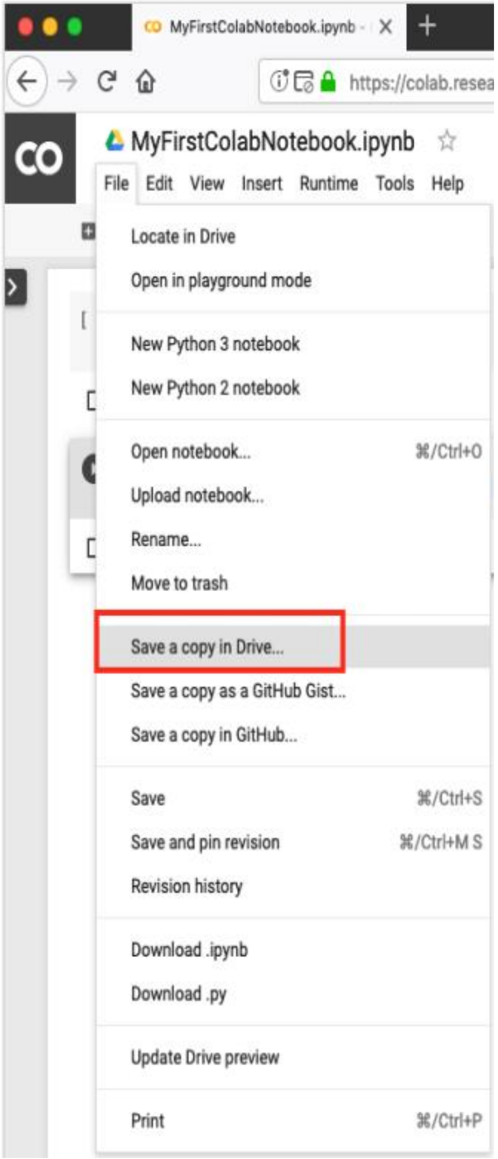
Markdown Examples

- Let us look into few examples of markup language syntax to demonstrate its capabilities. Type in the following text in the Text cell

```
This is **bold**.
This is *italic*.
This is ~strikethrough~.
```

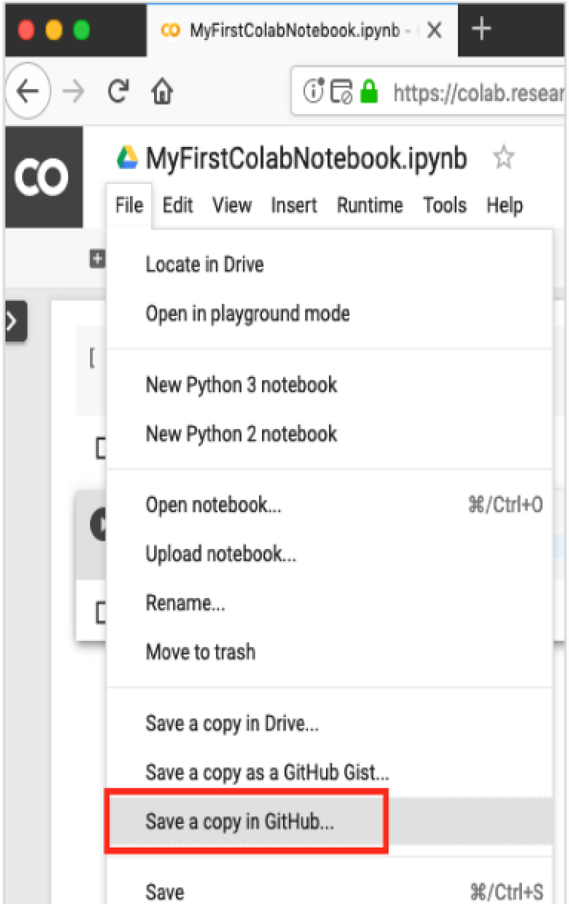
5. Google Colab – Saving Your Work

- Colab allows you to save your work to your Google Drive. To save your notebook, select the following menu options:

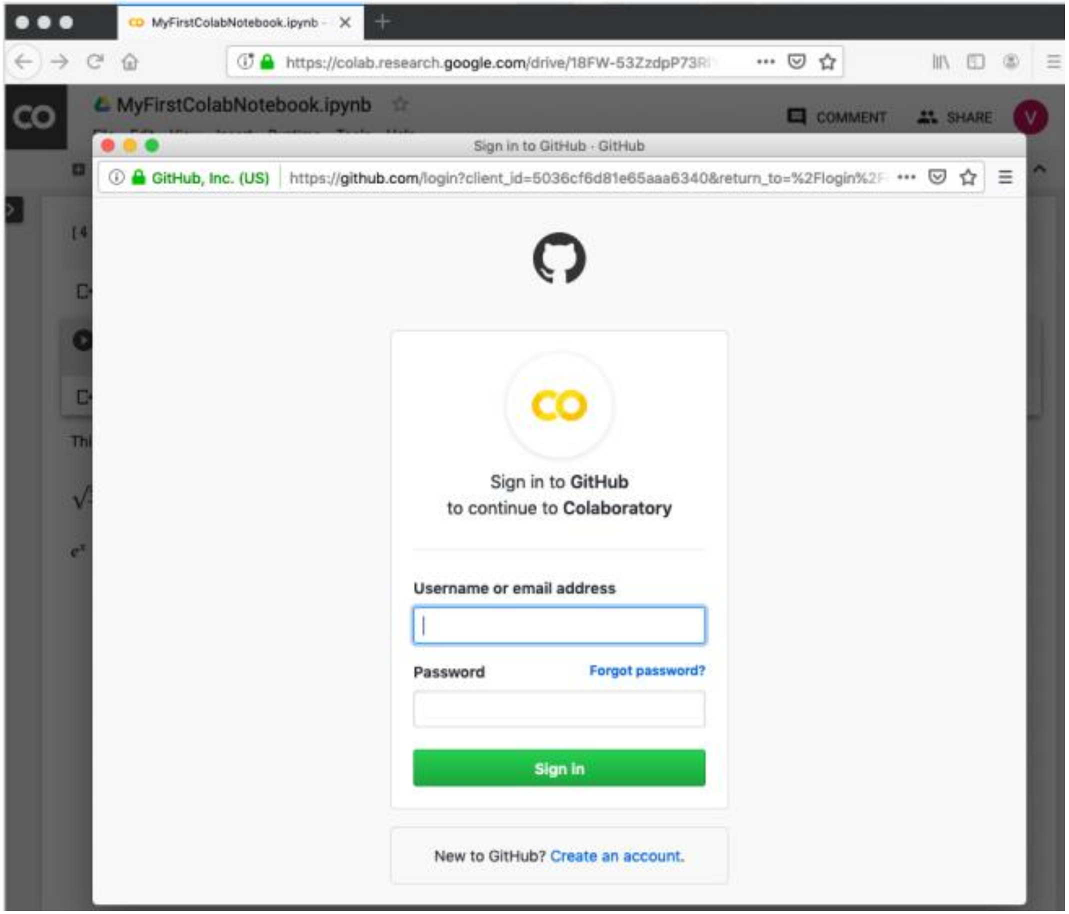


Saving to GitHub

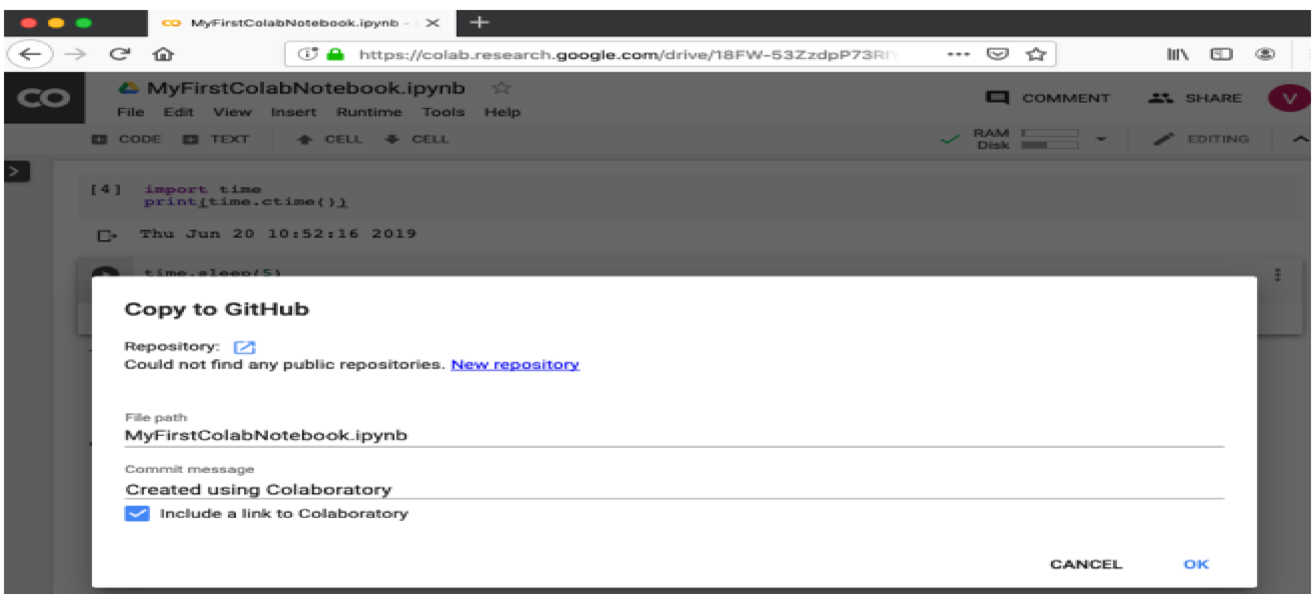
The menu selection is shown in the following screenshot for your quick reference:



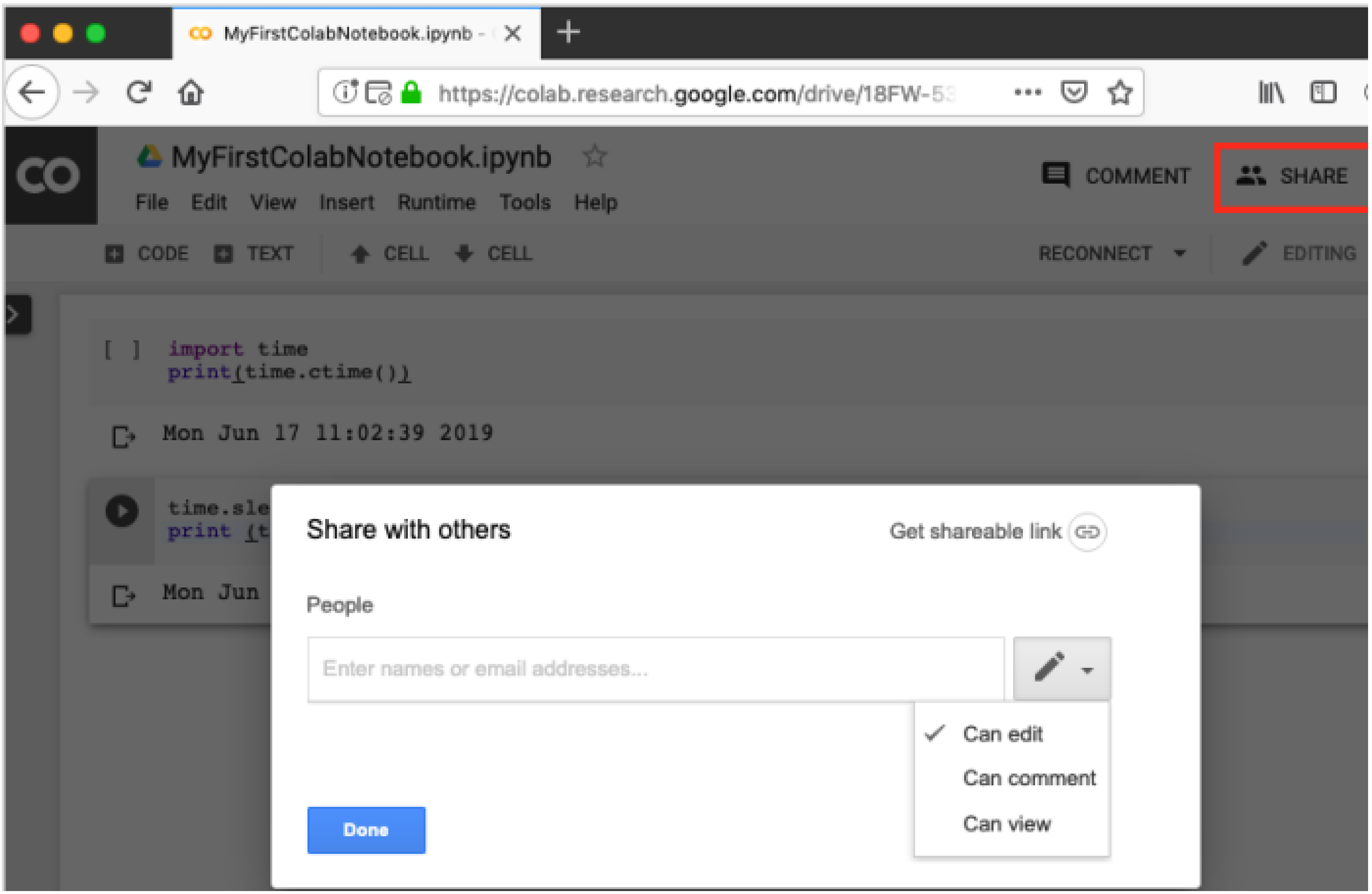
You will have to wait until you see the login screen to GitHub.



- Now, enter your credentials. If you do not have a repository, create a new one and save your project as shown in the screenshot below:



Google Colab – Sharing Notebook

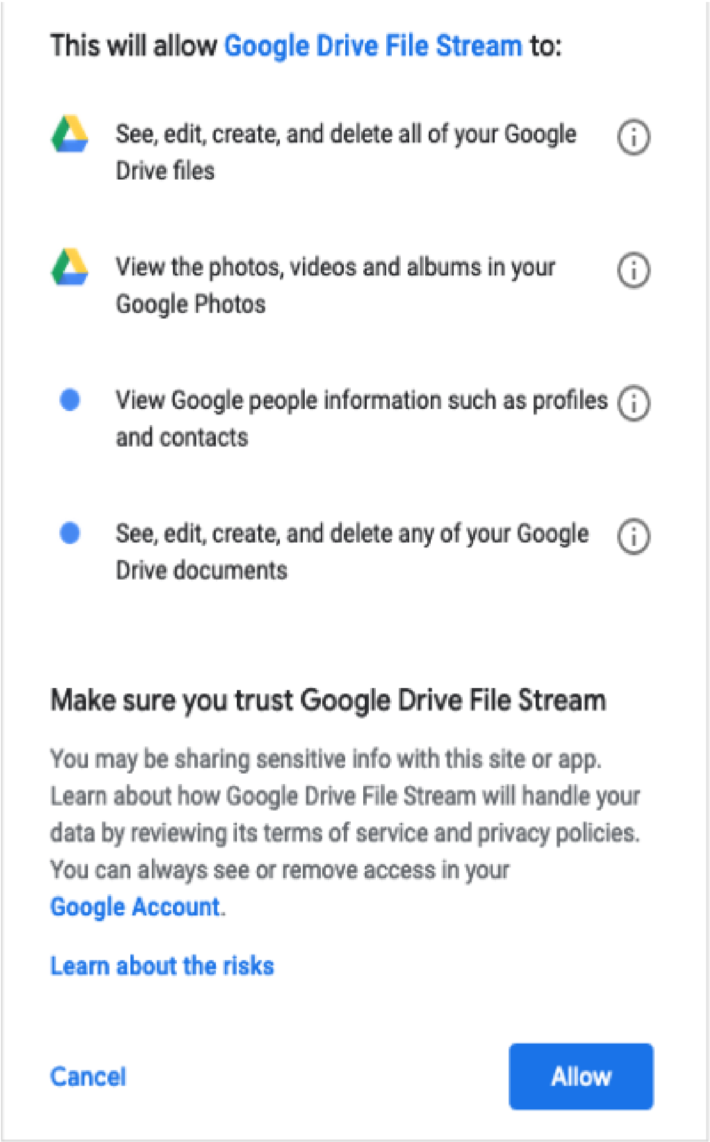


Mounting Drive

```
# Run this cell to mount your Google Drive.
from google.colab import drive
drive.mount('/content/drive')
```

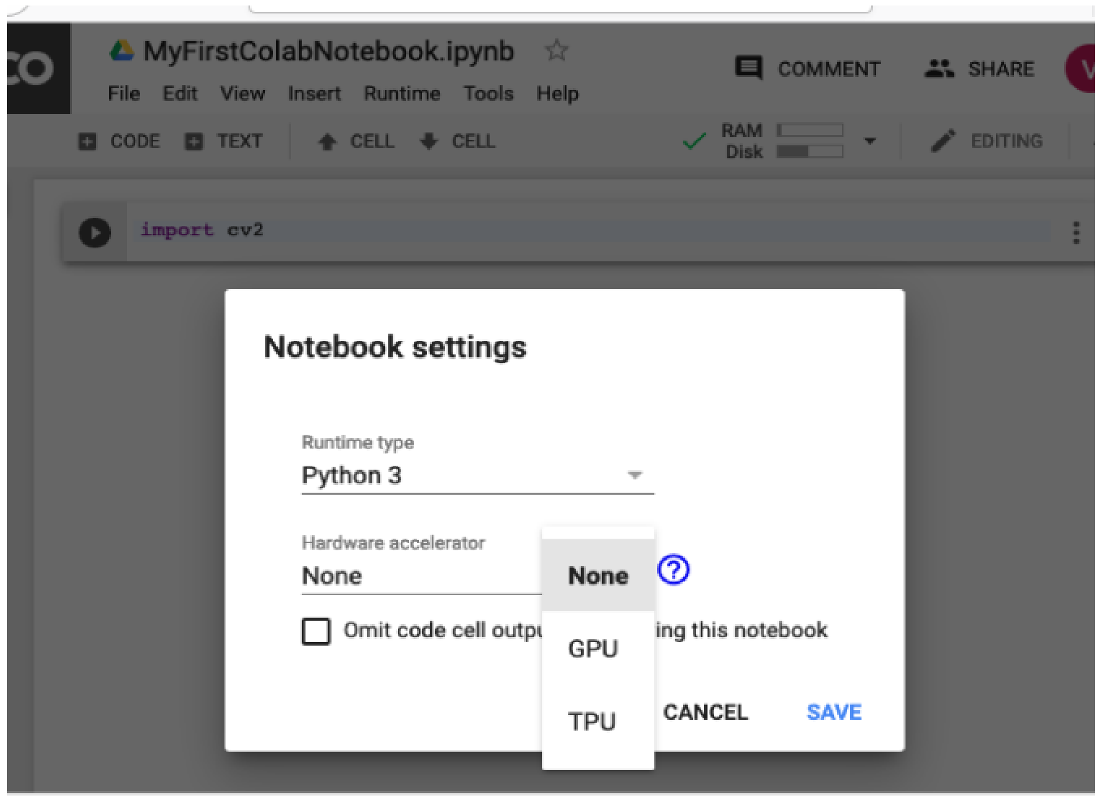
- If you run this code, you will be asked to enter the authentication code. The corresponding screen looks as shown below:

Google Drive File System



Enabling GPU

- To enable GPU in your notebook, select the following menu options:
- Runtime / Change runtime type



Basic Python Example

Example:

- print('Hello, World!')
- x = 10
- y = 5
- print(x + y)

Working with Libraries

Example:

```
import numpy as np
arr = np.array([1,2,3])
print(arr * 2)
```

Data Handling Example

Example:

```
import pandas as pd

data = {'Name':['Ali','Sara'],
'Age':[20,22]}

df = pd.DataFrame(data)

print(df)
```

Visualization Example

Example:

```
import matplotlib.pyplot as plt
x = [1,2,3]
y = [2,4,6]
plt.plot(x,y)
plt.show()
```

Using Google Drive

- from google.colab import drive
- drive.mount('/content/drive')
- Access your files directly from Drive

Advantages & Limitations

- Advantages:
- Free GPU
- Easy to use
- Limitations:
- Session timeout
- Limited RAM

Conclusion

Google Colab is a powerful tool for beginners and researchers.

Perfect for AI, ML, and data science projects.